

Evaluating Policy Gradient Methods and Domain Randomization for Sim-to-Real Robot Control

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I. INTRODUCTION

Reinforcement Learning has recently shown remarkable success in solving continuous control tasks, enabling agents to acquire manipulation and locomotion skills.

However, training these policies on real physical hardware is often unfeasible due to costs, mechanical wear and safety constraints. As a result of this, the standard procedure relies on training these agents in simulated environments before deploying them into the real world.

A major challenge in this procedure is the *reality gap*, the discrepancy between the simulated training environment and the actual world. This is often caused by unmodeled dynamics or sensor noise leading to a degradation of performance when moving from the simulation to the real world.

The first part of this work focuses on the `Hopper-v4` locomotion task, an environment where a one-legged robot must learn to jump forward without falling. Within this setting, we implement the vanilla REINFORCE algorithm from scratch and evaluate the critical role of baseline subtraction by testing both a zero baseline and a constant baseline.

Finally, we design and implement a one-step Actor-Critic architecture to show how bootstrapping and temporal difference learning can reduce gradient variance. This improves learning stability and speeds up policy convergence.

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II. RELATED WORK

III. METHODOLOGY

IV. EXPERIMENTS & RESULTS

A. Experimental Setup

The training on the `Hopper-v4` environment lasted for 5000 episodes, this number was empirically found to be sufficient for high-variance algorithms like REINFORCE in order to converge ensuring an exhaustive comparison.

All models in this environment were trained using the Adam optimizer with a fixed learning rate of 10^{-3} and a discount factor $\gamma = 0.99$.

Furthermore, to account the stochasticity of policy gradients methods, for each configuration we used 3 independent runs using different random initializers.

All the policies were tested on 50 test episodes.

B. Policy Gradient Analysis: REINFORCE

C. Variance Reduction: Actor-Critic vs. REINFORCE

V. CONCLUSIONS

REFERENCES



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